Statistical Modeling Project 2: EPA Air Data

MTH 4324/MTH 5324: Statistical Modeling

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Abstract

This study examined factors associated with poor air quality using 2022 Environmental Protection Agency (EPA) air quality data. Multiple datasets containing PM2.5, ozone, NO_2, temperature, humidity, and windspeed measurements were merged and aggregated by state and day to analyze trends in particulate pollution across the United States. Two response variables were modeled. The first was a binary indicator representing whether PM2.5 exceeded the World Health Organization (WHO) recommended threshold of 15 $\mu g/m^3$. The second response variable measured the number of PM2.5 exceedances within each grouped observation. Generalized linear models (GLMs) were used because the response variables were binary and count-based rather than continuous. Logistic regression with a logit link function was used to model PM2.5 threshold exceedance, while negative binomial regression with a log link function was used to model exceedance counts after overdispersion was identified in the count data. The results showed that PM2.5 exceedance behavior was influenced by a combination of environmental, seasonal, and geographic factors. Temperature, humidity, NO_2, quarter, and geographic division consistently contributed to model performance, while ozone showed little predictive value. Interaction terms involving temperature, humidity, quarter, and windspeed improved the overall model fit by capturing more complex atmospheric relationships. The Pacific and Mountain divisions showed the highest PM2.5 levels and exceedance counts, likely reflecting wildfire-related pollution patterns. Overall, the models successfully captured important air quality trends and demonstrated the usefulness of GLMs for environmental data analysis.

1 Introduction

Understanding how different factors influence real-world outcomes is an important part of data analysis. In many cases, outcomes are not just numerical data, but are also split into different categories or answer yes-or-no questions. This creates a challenge because standard linear regression models are not always appropriate for these types of responses. This problem matters because using the wrong model type can lead to inaccurate conclusions and poor predictions. Generalized Linear Models (GLMs) provide a better framework for handling binary and count data, allowing for more accurate modeling and interpretation. These models are widely used across fields such as public health, economics, and environmental science to better understand patterns and relationships in complex data. Some of these fields are evident in our first projects in this course, which explored COVID-19 mortality rates and temperature anomalies.

In this second project, we focus on EPA air quality data. We combine multiple datasets, including PM2.5, ozone, temperature, and other environmental factors, to provide a more comprehensive view of air quality. By merging these data sources, we can uncover deeper trends and better understand how different environmental variables interact.

The goal of this analysis is to model two response variables: whether a pollution threshold is exceeded (binary) and the number of exceedances (count). Key predictors include PM2.5 levels, ozone, NO_2, temperature, humidity, wind, location, and day type. By applying GLMs, we can determine how these factors influence both the likelihood and frequency of poor air quality events.

2 Data Description

2.1 Data Source and Structure

Data came from the public access Environmental Protection Agency (EPA) database and included observations from 2022. Each observation represented one collection per hour with different stations. Prior to cleaning and preprocessing, the combined dataset contained approximately 2.9 million observations. After cleaning, aggregation, and removal of incomplete records, the final dataset contained approximately 16.9 thousand observations.

The primary air quality measurement used in this study was particulate matter 2.5 micrometers and smaller (PM2.5), reported in micrograms per cubic meter ($\mu g/m^3$). PM2.5 represents very small airborne particles commonly associated with smog, combustion, and wildfire smoke. Because these particles are small enough to enter the lungs and bloodstream, elevated PM2.5 levels are strongly associated with respiratory and cardiovascular health risks.

2.2 Response Variables

2.2.1 Response Variable 1: PM2.5 Threshold Exceedance (Binary)

The first response variable was a binary indicator representing whether PM2.5 exceeded the World Health Organization (WHO) recommended public health threshold of 15 μg/m³. While the EPA standard is higher at 35 μg/m³, the WHO threshold was selected because it reflects stricter public health recommendations and identifies air quality levels associated with increased respiratory and health concerns. The binary response variable was either a value of 1, representing a "bad air day," while a value of 0 represented acceptable air quality conditions.

2.2.2 Response Variable 2: Number of PM2.5 Exceedances (Count)

The second response variable measured the number of PM2.5 exceedances within each aggregated state-day observation. After averaging and grouping the data, an additional variable was created to count how many times PM2.5 exceeded the 15 μg/m³ threshold. This variable was used for count-based generalized linear modeling.

2.3 Predictor Variables

2.3.1 Environmental Predictors

Nitrogen dioxide (NO_2), reported in parts per billion, was included because it contributes to atmospheric reactions that form particulate matter, especially PM2.5. Windspeed, reported in knots, was included because stronger winds can disperse pollutants and reduce particulate concentrations. Temperature, reported in degrees Fahrenheit, was included because atmospheric chemical reactions are highly temperature-dependent. Humidity, reported as relative percent, was included because moisture and atmospheric conditions can influence pollutant behavior. Ozone, reported in parts per million, was also included because it forms through photochemical reactions associated with combustion and transportation emissions.

2.3.2 Geographic and Seasonal Predictors

Geographic predictors were demonstrated by the U.S. Census divisions. Divisions such as Pacific, Mountain, South Atlantic, and New England were used instead of broader regional categories in order to show more detailed variation in air quality patterns. Seasonal variation was represented using quarters (Q1–Q4), allowing the analysis to demonstrate changes associated with the seasons.

2.4 Data Cleaning and Preprocessing

Before conducting any analysis, the dataset required substantial cleaning and preprocessing to ensure consistency and reliability. Because the data was created by merging multiple environmental datasets, there were many missing values, inconsistencies in measurement types, and variables that were not useful for modeling. To address this, the dataset was narrowed to include only the most relevant pollution, weather, and geographic variables.

2.4.1 Handling Inconsistent Measurements

One major issue involved the inconsistent recording of environmental measurements. Within the temperature dataset, some observations represented humidity while others represented dew point values. A similar issue occurred within the wind dataset. Approximately half of the observations represented wind direction, while the remaining observations represented wind speed. Excel pivot tables were used to reorganize and separate the measurements, retaining only the relevant humidity and wind speed values while removing dew point and wind direction.

2.4.2 Data Transformation

After selecting the appropriate variables, all continuous variables, including PM2.5, ozone, NO_2, humidity, temperature, and windspeed, were converted to numeric values. Categorical variables such as state, division, and quarter were treated as factors for modeling purposes.

The data was aggregated by state and by day because, without aggregation, there were virtually no complete observations. Within each location, variables such as temperature, humidity, ozone, NO_2, and PM2.5 were averaged. U.S. Census divisions were also included to provide more detailed geographic structure than broader regional groupings. Several additional variables were created to better capture air quality behavior. These included a binary indicator for whether PM2.5 exceeded 15 $\mu g/m^3$, as well as summary measures such as exceedance counts, maximum PM2.5, and mean PM2.5 values within each grouping.

2.4.3 Missing Values

Finally, after all transformations and aggregations were completed, rows containing missing values were removed. This ensured that the final dataset used for generalized linear modeling was complete, consistent, and suitable for statistical analysis.

3 Exploratory Analysis

3.1 Distribution of PM2.5

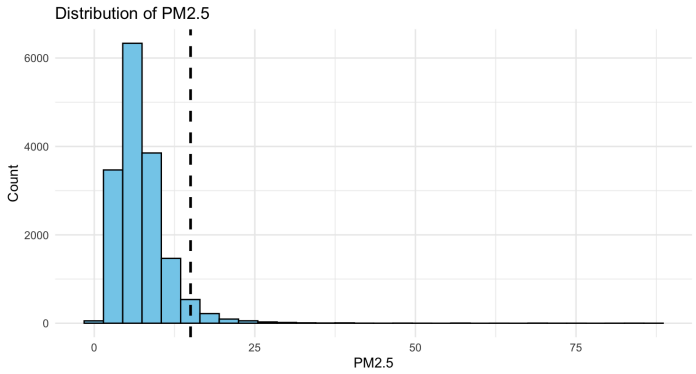


Figure 1: *Frequency distribution of PM2.5 values, with the vertical dashed line indicating the 15 µg/m³ threshold used to determine a "bad air day."*

The first step in the exploratory analysis was to examine the distribution of PM2.5 for all of our data. As shown in Figure 1, PM2.5 values are right-skewed, with most observations concentrated at lower levels and a long tail extending toward higher values. However, the distribution is not centered at zero, and the highest concentration of observations occurs at a small positive value instead. This creates a slightly curved shape rather than a sharp spike at zero. This pattern is expected in real-world air quality data, since PM2.5 levels are almost never truly zero. Even in relatively clean environments, there are always background levels of particulate matter due to natural sources such as dust, pollen, and sea salt, as well as low levels of human activity like transportation and industry. As a result, the distribution naturally centers around a low baseline instead of zero.

Most observations fall below this threshold, indicating that exceedance events are less common but still important to model due to their potential impact on public health. Additionally, the skewed shape and presence of higher-value outliers suggest that standard linear modeling assumptions, such as normality, may not hold. Overall, this initial analysis shows that PM2.5 is not normally distributed, has a natural lower bound above zero, and includes occasional extreme values.

3.2 PM2.5 by Local Climate and Season

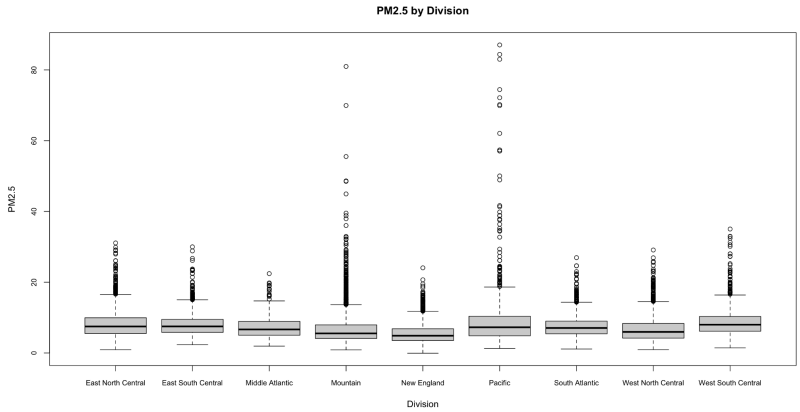


Figure 2: *Boxplot of PM2.5 concentrations across U.S. Census Divisions.*

Some divisions of the country had significantly higher PM2.5 levels than others, indicating that the region does affect PM2.5 levels (Figure 2). New England consistently shows some of the lowest PM2.5 levels with a tight distribution, while the Mountain and Pacific divisions showed much larger variation.

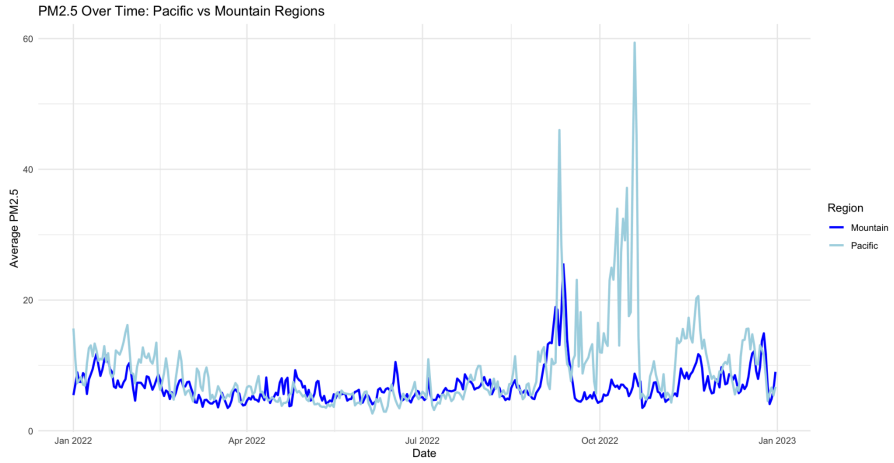


Figure 3: *Time series of averaged PM2.5 concentrations in the Mountain and Pacific divisions.*

The Pacific and Mountain regions stand out with notably higher PM2.5 levels, a wider spread, and more extreme outliers. This indicates that these regions not only experience higher average pollution levels but also more frequent spikes in PM2.5. There was a clear increase in PM2.5 levels during the fall months, with a noticeable peak in October (Figure 3). This pattern aligns closely with the wildfire season in the western US. During this time, large wildfires release significant amounts of particulate matter into the atmosphere, leading to sharp increases in PM2.5 concentrations.

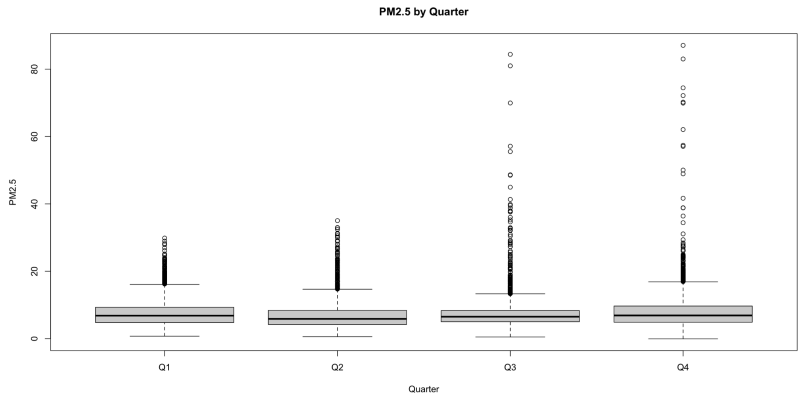


Figure 4: *Boxplot of PM2.5 concentrations by quarter.*

While the median PM2.5 levels remain relatively similar across all four quarters, there is a noticeable increase in extreme values in Q3 and Q4 (Figure 4). This pattern supports earlier observations from the Pacific and Mountain region analysis, which showed that PM2.5 levels peak in the fall months due to wildfire activity. Since Q4 includes October, November, and December, the increased spread and extreme values in this quarter are consistent with the timing of wildfire season. Similarly, Q3 shows increased variability, suggesting a buildup toward peak wildfire conditions.

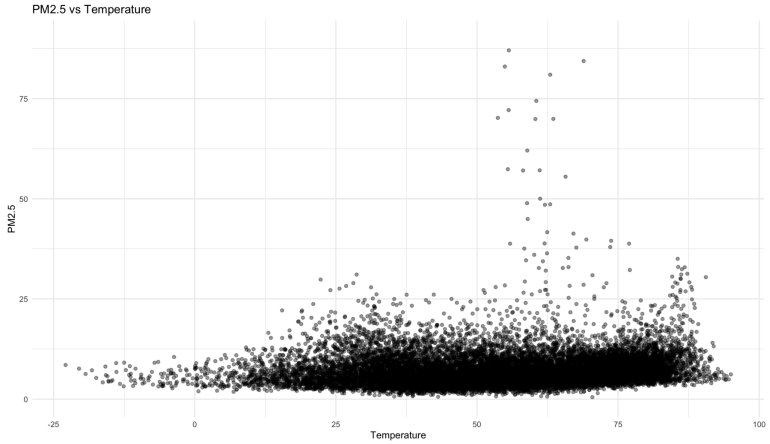


Figure 5: *Scatter plot of PM2.5 concentrations versus temperature.*

PM2.5 versus temperature shows a wide spread of values, with no strong linear relationship between the two variables (Figure 5). Most observations are concentrated at lower PM2.5 levels across all temperatures, whereas higher PM2.5 values are more scattered and occur across a wider range of temperatures. This suggests that temperature alone is not a strong direct predictor of PM2.5.

A notable feature of the graph is the concentration of extreme outliers in the temperature range of approximately 50 to 75 degrees. Within this range, several observations show very high

PM2.5 values compared to the rest of the data. This suggests that extreme pollution events are more likely to occur under moderate temperatures than at very low or very high temperatures. These patterns further support the idea that the relationship between temperature and PM2.5 is complex and likely influenced by other factors, such as seasonal effects and environmental conditions like wildfires.

3.3 PM2.5 vs Humidity

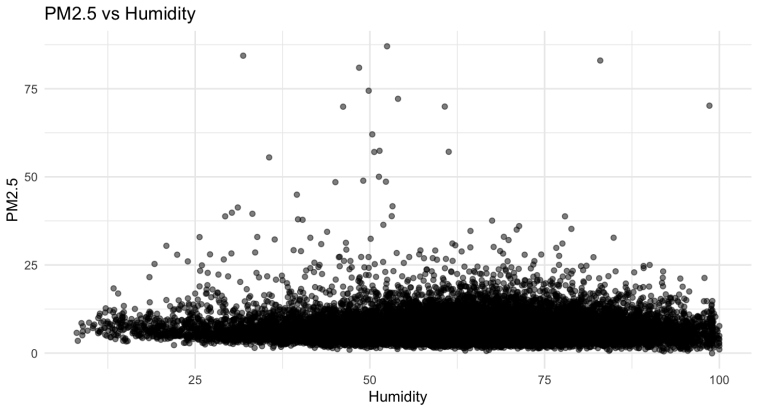


Figure 6: *Scatter plot of PM2.5 concentrations versus humidity.*

Most observations are clustered at lower PM2.5 levels across all humidity values, but several higher outliers occur, especially around moderate humidity levels (Figure 6). There is also a slightly curved pattern, where PM2.5 values appear somewhat higher at mid-range humidity levels and lower at very low or very high humidity, suggesting that humidity may have a nonlinear effect on PM2.5.

When considered alongside temperature, humidity likely plays a role in shaping environmental conditions that affect air quality. Since both humidity and temperature are related to weather and atmospheric conditions, their effects on PM2.5 may depend on each other rather than acting independently. Because of this, we think humidity may be more useful when included as an interaction with something like temperature, rather than as a standalone predictor. This would allow the model to better capture how combined weather conditions influence PM2.5 levels.

3.4 PM2.5 Relation to NO_2 and Windspeed

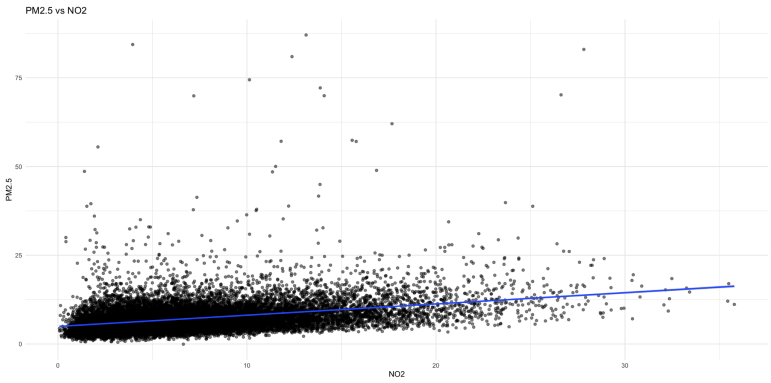


Figure 7: *Scatter plot of PM2.5 concentrations versus NO_2, with the blue line indicating the best-fit regression line and a positive correlation between variables.*

PM2.5 versus NO_2 shows a slight positive relationship, and PM2.5 tends to increase as NO_2 increases (Figure 7). While the relationship is not extremely strong, the upward trend line suggests that higher NO_2 levels are generally associated with higher PM2.5 levels.

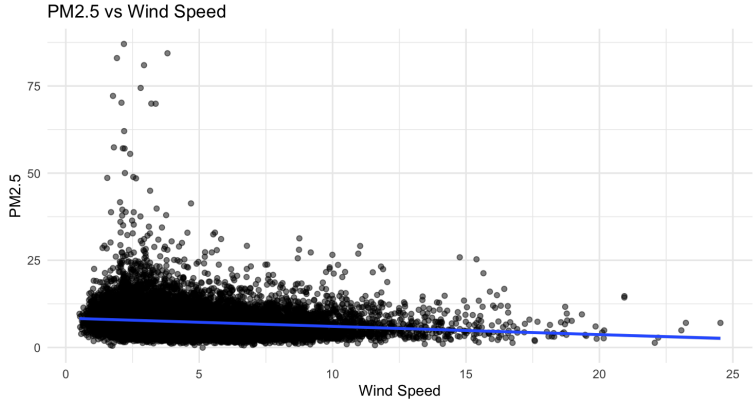


Figure 8: *Scatter plot of PM2.5 concentrations versus wind speed (knots), with the blue line indicating the best-fit regression line and a negative correlation between variables.*

PM2.5 versus wind speed shows a slight negative relationship (Figure 8). As wind speed increases, PM2.5 levels tend to decrease. This makes sense physically: higher wind speeds help disperse pollutants in the air, reducing particle concentrations. Similar to the NO_2 graph, there is still a wide spread of values and some outliers, but the general downward trend suggests that wind speed may also be an important predictor.

It may be useful to consider an interaction between NO_2 and wind speed in the model. For example, the effect of NO_2 on PM2.5 may be stronger when wind speeds are low, since pollutants are less likely to scatter. On the other hand, at higher wind speeds, the impact of NO_2 on PM2.5 may be reduced due to increased dispersion. Including an interaction term between

NO_2 and wind speed would allow the model to capture how these variables work together to influence PM2.5, rather than assuming their effects are completely separate.

3.5 PM 2.5 vs Ozone

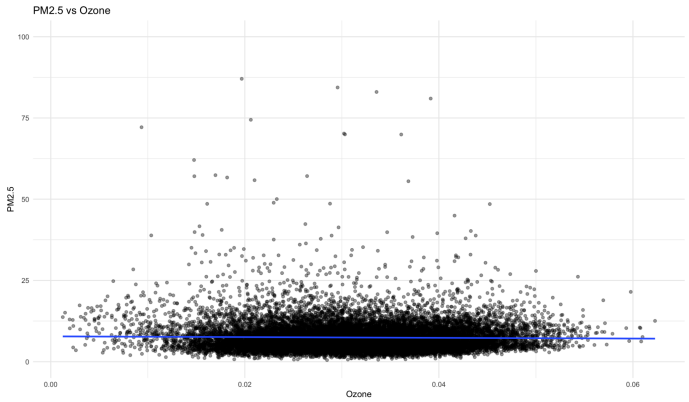


Figure 9: *Scatter plot of PM2.5 concentrations versus ozone, with the blue line indicating the best-fit regression line and no correlation between variables.*

The scatterplot of PM2.5 versus ozone shows no clear relationship between the two variables. The data points are widely scattered, and there is no noticeable upward or downward trend. Unlike other variables such as NO_2 or wind speed, ozone does not appear to have a consistent effect on PM2.5 levels. In atmospheric chemistry, ozone and PM2.5 have differing correlations. PM2.5 can be produced through chemical reactions in the environment, but also through the direct release from combustion or anthropogenic actions. Ozone, on the other hand, is mainly directly a production of photochemical reactions. This means that ozone is a sign that chemical reactions are occurring that could produce increased PM2.5 levels, yet is not a perfect predictor. Because of this lack of pattern, ozone is unlikely to be a strong predictor of PM2.5 on its own.

3.6 Correlation Heatmap

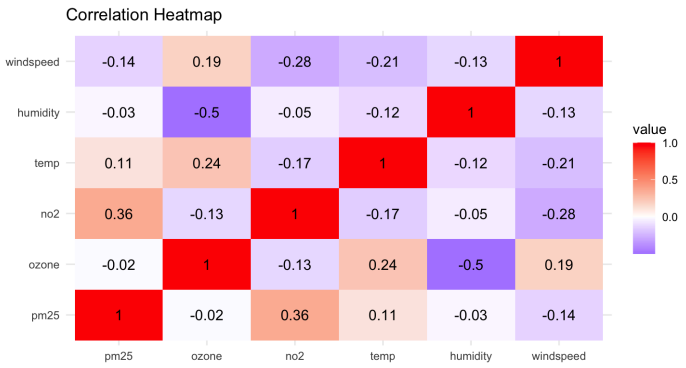


Figure 10: *Correlation heatmap of variables, where red indicates stronger positive correlations, blue indicates stronger negative correlations, and white represents weak or negligible relationships.*

The correlation heatmap shows the strength and direction of the relationships between variables, with Pearson's correlation values ranging from -1 to 1 (Figure 10). Values closer to 1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values near 0 indicate little to no relationship.

Looking at PM2.5, most correlations are relatively small, suggesting that no single numeric variable explains PM2.5 on its own. The largest value with NO_2 (about 0.36), indicating a moderate positive relationship that matches the upward trend in the scatterplot. Wind speed shows a slight negative correlation (around -0.14), consistent with the idea that higher wind speeds help reduce PM2.5 levels. Temperature and humidity both show very weak correlations with PM2.5, reinforcing what was seen earlier: that they do not have strong individual relationships but may still play a role in combination with other variables. Ozone shows almost no correlation with PM2.5 (-0.02), suggesting it does not have a meaningful relationship with pollution levels.

The heatmap also shows relationships among predictors, such as the moderate negative correlation between ozone and humidity (around -0.5), suggesting that these variables are related and should not both be included in order to avoid multicollinearity. Overall, the heatmap reinforces the patterns observed in the individual graphs and supports the idea that PM2.5 is influenced by multiple factors rather than a single strong predictor.

4 Methodology and Model Selection

4.1 Generalized Linear Models

Because the response variables in this study were binary and count-based, generalized linear models GLMs were used instead of standard linear regression. GLMs allow response variables to follow distributions other than a standard normal distribution and use link functions to model relationships between predictors and responses.

Two final models were developed: a logistic regression model for PM2.5 threshold exceedance and a negative binomial regression model for exceedance counts. Multiple candidate models were tested throughout the modeling process using different combinations of pollution, weather, seasonal, and geographic predictors. Model selection was primarily based on Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), overall fit, and interpretability.

4.2 Logistic Regression Model

The first response variable represented whether PM2.5 exceeded 15 µg/m³. Since the response only had two possible outcomes, logistic regression was used with a logit link function. The logit link models probabilities on the log-odds scale and ensures predicted probabilities remain between 0 and 1.

The binary response variable was defined as:

$$Y_i = \begin{cases} 1, & \text{if PM2.5} > 15 \\ 0, & \text{otherwise} \end{cases}$$

To develop the final model, several candidate logistic regression models were compared using different combinations of predictors and interaction terms. Earlier models focused primarily on pollution variables, while later models incorporated weather variables, seasonal effects, geographic divisions, and interaction terms. Model performance was evaluated using AIC and BIC values, where lower values indicated a stronger fit relative to model complexity.

Model	Variables	AIC	BIC
m1	pm25_max	4.04	19.42
m2	ozone + no2	16547.43	16570.50
m3	ozone + no2 + temp + humidity + windspeed	16476.95	16523.09
m4	ozone + no2 + temp + humidity + windspeed + division + quarter	15085.23	15215.97
m5	ozone + no2 + windspeed + quarter + temp*ozone + temp*humidity + division	14864.62	15010.75
m6	ozone*humidity + temp*quarter + no2 + division	14730.10	14883.92
m7	temp*humidity + temp*quarter + no2 + division	14751.89	14898.02
m8	ozone*humidity + temp*quarter + windspeed + no2 + division	14707.91	14869.41
m9	humidity*temp + temp*quarter + windspeed*no2 + division	14675.73	14837.24

Table 1: *Model selection diagnostics showing AIC and BIC values, with the lowest values at the bottom corresponding to the final selected model.*

Models containing only pollution variables generally produced higher AIC and BIC values than models incorporating weather and regional effects (Table 1). Interaction terms further improved fit statistics, particularly the interactions between temperature and humidity, temperature and quarter, and wind speed and NO_2.

The final logistic regression model produced the lowest overall AIC and BIC values while remaining interpretable and environmentally meaningful. The final model was:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1(\text{humidity}) + \beta_2(\text{temp}) + \beta_3(\text{humidity} \times \text{temp}) + \beta_4(\text{temp} \times \text{quarter}) + \beta_5(\text{windspeed} \times \text{NO}_2) + \beta_6(\text{division})$$

Confusion matrices were generated across classification thresholds from 0.10 to 0.90. A threshold of 0.30 was selected, yielding an accuracy of ~75% (Table 2). Although some thresholds produced higher overall accuracy, they resulted in increased false positives and false negatives. Predicted probabilities from the final model were then compared to observed exceedance outcomes to assess classification performance.

Metric	Value
Accuracy	0.753
Sensitivity	0.632
Specificity	0.795
Cutoff Value	0.300

Table 2: *Confusion matrix with 0.30 selected as the optimal threshold (~75% accuracy) balancing false positives and false negatives.*

4.3 Negative Binomial Model

The second response variable represented the number of PM2.5 exceedances within each observation. A Poisson model was initially considered, but data was overdispered, where the variance exceeded the mean. Because this violated Poisson assumptions, a negative binomial model was selected instead. The log link function was used because the response variable consisted of positive count data, and the transformation allows predictors to affect the log expected exceedance count.

Multiple negative binomial models were evaluated using different combinations of predictors and interaction terms.

Model	Variables	AIC	BIC
c1	pm25_max	40379.07	40402.14
c2	ozone + no2	45721.32	45752.08
c3	ozone + no2 + temp + humidity + windspeed	45518.67	45572.51
c4	ozone + no2 + temp + humidity + windspeed + division + quarter	44544.26	44682.70
c5	ozone + no2 + windspeed + quarter + temp*ozone + temp*humidity + division	44352.86	44506.67
c6	ozone*humidity + temp*quarter + no2 + division	43966.55	44128.06
c7	temp*humidity + temp*quarter + no2 + division	43940.91	44094.73
c8	ozone*humidity + temp*quarter + windspeed + no2 + division	43933.03	44094.53
c9	humidity*temp + temp*quarter + windspeed*no2 + division	43905.39	44074.59

Table 3: *Model selection diagnostics showing AIC and BIC values, with the lowest values at the bottom corresponding to the final selected model.*

Simpler count models produced higher information criteria values and failed to capture the full variability in exceedance counts (Table 3). Including interaction terms reduced both AIC and BIC and improved overall model fit. The final negative binomial model was:

$$\log(E[Y_i]) = \beta_0 + \beta_1(\text{humidity}) + \beta_2(\text{temp}) + \beta_3(\text{humidity} \times \text{temp}) + \beta_4(\text{temp} \times \text{quarter}) + \beta_5(\text{windspeed} \times \text{NO}_2) + \beta_6(\text{division})$$

4.4 Interaction Terms

The interaction between temperature and humidity was included because temperature affects how moisture behaves in the atmosphere and can influence the formation and concentration of particulate matter. Moderate to high humidity combined with warmer temperatures may create conditions that allow pollutants to remain suspended in the air more easily.

The interaction between temperature and quarter was included to represent seasonal differences in temperature effects throughout the year. Seasonal changes affect atmospheric chemistry, wildfire activity, and pollutant accumulation, causing temperature effects to vary across quarters.

The interaction between NO_2 and windspeed was included because wind directly affects pollutant dispersion. Higher NO_2 levels are generally associated with increased particulate formation, but stronger winds can disperse pollutants and reduce buildup.

4.5 Model Assumptions

Several assumptions were considered for both final models. Since the data was aggregated by state and day, observations were treated as conditionally independent. The logistic regression model assumes that the predictors are linearly related to the log-odds of PM2.5 exceedance, while the negative binomial model assumes the response is count-based and allows the variance to be larger than the mean. This is why the negative binomial model was used instead of a Poisson model.

Correlation heatmaps and exploratory analysis were also used to check for possible multicollinearity. Ozone was removed during model selection because it did not contribute meaningful information and was correlated with other atmospheric variables, which could weaken model stability.

5 Results

5.1 Logistic Regression Results

The final logistic regression model was used to estimate the probability that PM2.5 exceeded the WHO threshold of 15 µg/m³. Several predictors were statistically significant, including humidity, temperature, quarter, division, and the interaction terms included in the final model (Table 4).

Predictor	Estimate	Std_Error	z_value	p_value
(Intercept)	-4.1995	0.3440	-12.208	0.00000
humidity	0.0273	0.0045	6.128	0.00000
temp	0.0505	0.0056	9.029	0.00000
quarterQ2	-5.6119	0.3420	-16.407	0.00000
quarterQ3	-2.4170	0.4919	-4.914	0.00000
quarterQ4	0.4160	0.1596	2.606	0.00916
windspeed	0.0512	0.0157	3.249	0.00116
no2	0.1902	0.0092	20.775	0.00000
divisionEast South Central	-0.5153	0.0914	-5.636	0.00000
divisionMiddle Atlantic	0.0649	0.1020	0.637	0.52429
divisionMountain	-0.2888	0.0805	-3.586	0.00034
divisionNew England	-1.8711	0.1101	-16.994	0.00000
divisionPacific	0.8698	0.0918	9.471	0.00000
divisionSouth Atlantic	-1.2426	0.0878	-14.156	0.00000
divisionWest North Central	-0.3765	0.0864	-4.357	0.00001
divisionWest South Central	0.0404	0.0919	0.439	0.66050
humidity:temp	-0.0005	0.0001	-7.190	0.00000
temp:quarterQ2	0.0701	0.0053	13.338	0.00000
temp:quarterQ3	0.0177	0.0068	2.605	0.00919
temp:quarterQ4	-0.0086	0.0035	-2.429	0.01513
windspeed:no2	-0.0168	0.0021	-7.890	0.00000

Table 4: *Final logit model coefficient estimates, standard errors, z-values, and p-values.*

Temperature and humidity both showed important effects in the final model, particularly through their interaction term. This suggests that atmospheric conditions influence PM2.5 together rather than independently. Geographic divisions such as the Pacific and Mountain regions also showed increased probabilities of exceedance compared to other regions, supporting patterns observed during exploratory analysis.

5.2 Negative Binomial Results

The final negative binomial model was used to estimate the number of PM2.5 exceedances within each grouped observation. The model showed that environmental, seasonal, and geographic variables all contributed to variation in exceedance counts (Table 5).

Predictor	Estimate	Std_Error	z_value	p_value
(Intercept)	-3.7079	0.4051	-9.154	0.00000
humidity	0.0424	0.0053	7.991	0.00000
temp	0.0794	0.0066	11.958	0.00000
quarterQ2	-5.7907	0.3227	-17.945	0.00000
quarterQ3	-3.0859	0.5223	-5.909	0.00000
quarterQ4	0.4753	0.1945	2.444	0.01453
windspeed	0.0286	0.0177	1.619	0.10540
no2	0.1853	0.0104	17.749	0.00000
divisionEast South Central	-0.8608	0.1103	-7.802	0.00000
divisionMiddle Atlantic	0.1510	0.1282	1.178	0.23875
divisionMountain	-0.4694	0.0984	-4.770	0.00000
divisionNew England	-2.6751	0.1135	-23.572	0.00000
divisionPacific	0.4734	0.1147	4.125	0.00004
divisionSouth Atlantic	-1.6749	0.1018	-16.459	0.00000
divisionWest North Central	-0.6079	0.1025	-5.933	0.00000
divisionWest South Central	-0.4115	0.1177	-3.495	0.00047
humidity:temp	-0.0008	0.0001	-8.907	0.00000
temp:quarterQ2	0.0719	0.0054	13.393	0.00000
temp:quarterQ3	0.0237	0.0074	3.219	0.00129
temp:quarterQ4	-0.0145	0.0043	-3.360	0.00078
windspeed:no2	-0.0133	0.0025	-5.260	0.00000

Table 5: *Final negative binomial model coefficient estimates, standard errors, z-values, and p-values.*

Similar to the logistic model, interaction terms involving temperature, humidity, quarter, and windspeed improved the model's ability to capture changing atmospheric conditions. Western regions generally showed higher exceedance counts than eastern regions, likely reflecting the influence of wildfire activity and regional climate differences.

6 Diagnostics and Validation

6.1 Residual Analysis

Residual plots were examined for both the logistic and negative binomial models to evaluate overall fit and identify unusual observations.

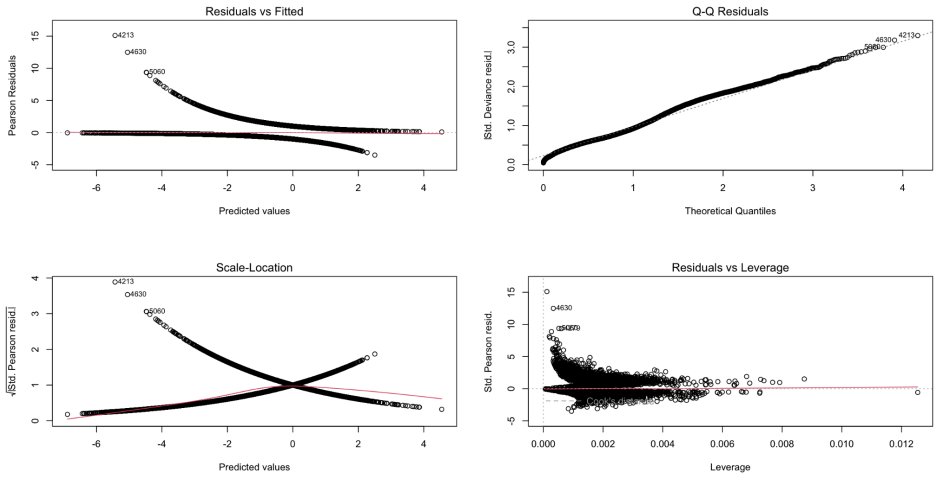


Figure 11: *Diagnostic plots for the logistic regression model, including residual vs fitted, Q–Q, scale-location, and residuals versus leverage plots.*

For the logistic regression model, the residual plots showed some curvature, which is expected with a binary response variable because the outcome can only be 0 or 1 (Figure 11). The Q-Q plot showed some deviation in the tails, suggesting that some observations were harder for the model to fit. The residuals versus leverage plot showed that most points had low leverage, meaning most observations did not overly influence the model. Overall, the diagnostics suggest that the logistic model is not perfect, but it is reasonable for modeling the probability of PM2.5 exceedance.

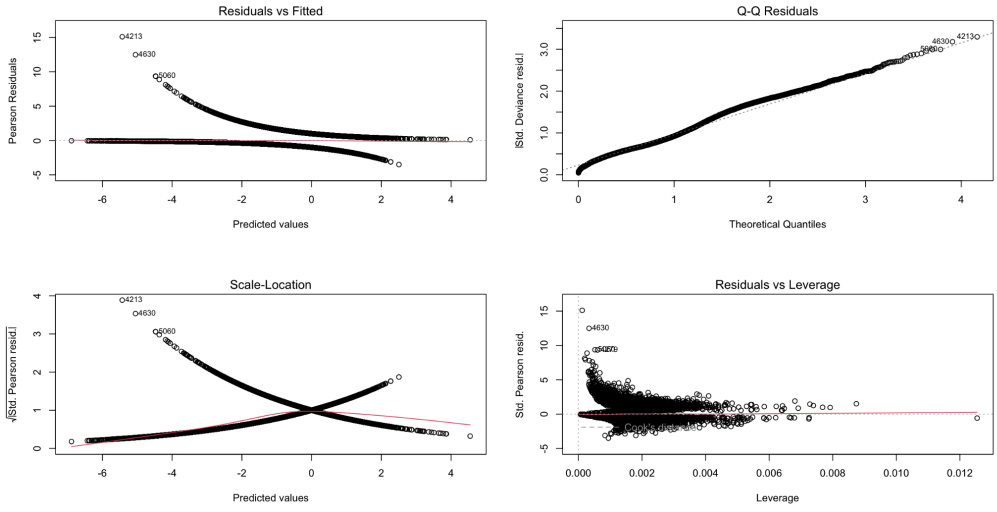


Figure 12: *Diagnostic plots for the negative binomial model, including residual vs fitted, Q–Q, scale-location, and residuals versus leverage plots.*

For the negative binomial model, the residual plots showed a wider spread than the logistic model, which makes sense because the count response has more variation (Figure 12). The Q-Q plot again showed some tail deviation, indicating that extreme exceedance counts were more difficult to capture. However, the residuals were mostly centered around zero, and the leverage plot showed that most observations were not highly influential. Since environmental data naturally contains extreme pollution events, these patterns were expected.

As a whole, the diagnostic plots show that both models captured the main structure of the data while still reflecting the natural variability of air quality measurements. The models are not perfect, but they provide a reasonable fit for explaining both the likelihood and number of PM2.5 exceedances.

6.2 Goodness-of-Fit and Overdispersion

Statistic	Value
AIC	14675.73
BIC	14837.24
Deviance	14633.73
Null Deviance	18322.95
Residual Deviance	14633.73

Table 6: *Final logit model diagnostics including AIC, BIC, deviance, null deviance, and residual deviance (right)*

Statistic	Value
AIC	43905.39
BIC	44074.59
Deviance	8952.99
Overdispersion Statistic	1.56

Table 7: *Final negative binomial model diagnostics including AIC, BIC, deviance, and overdispersion statistic.*

Model fit was evaluated using AIC, BIC, deviance, and residual diagnostics. For the logit model, AIC and BIC were relatively lowest, with acceptable deviance. For the count response variable, overdispersion checks showed that the variance substantially exceeded the mean, violating Poisson assumptions and supporting the use of a negative binomial model.

6.3 Validation

Prediction accuracy was evaluated using confusion matrices and classification metrics for the logistic regression model. A cutoff value of 0.3 produced the strongest balance between

sensitivity and overall accuracy. The final models successfully captured the major environmental patterns present in the dataset while maintaining reasonable predictive performance.

7 Discussion

The results of this study suggest that PM2.5 levels are influenced by a combination of weather, pollution, geographic, and seasonal factors rather than a single dominant predictor. Regional differences played a particularly important role, with the Pacific and Mountain divisions consistently showing higher PM2.5 levels and exceedance counts than other regions. These findings align closely with observed wildfire patterns in the western United States.

Interaction terms also contributed substantially to model performance. Temperature and humidity together appeared to influence atmospheric conditions differently than either variable alone, while the interaction between NO_2 and wind speed reflected how pollution dispersion changes under varying wind conditions.

One notable finding was the weak relationship between ozone and PM2.5. Although ozone is often associated with atmospheric pollution, exploratory analysis and model selection showed that it did not meaningfully improve prediction performance in this dataset.

There were several limitations to the analysis. Environmental data naturally contain extreme observations and variability, which can make perfect prediction difficult. In addition, aggregation by state and day reduces noise but may also hide smaller local patterns. Future work could incorporate additional environmental predictors, spatial modeling approaches, or finer geographic resolution to improve prediction accuracy.

8 Conclusion

This study used GLMs to examine factors associated with PM2.5 exceedance behavior across the United States. Logistic regression was used to model whether PM2.5 exceeded the WHO threshold of 15 $\mu g/m^3$, while negative binomial regression was used to model exceedance counts per state per division.

The results showed that environmental conditions, seasonal effects, and geographic location all influenced PM2.5 behavior. Temperature, humidity, NO_2, and regional divisions consistently contributed to model performance, while ozone showed little predictive value. Interaction terms improved both models by allowing environmental relationships to vary under different atmospheric conditions.

Overall, the final models successfully captured important air quality patterns and demonstrated how GLMs can be applied to complex environmental datasets. Future research could explore additional predictors or higher-resolution data to improve model performance.

9 References

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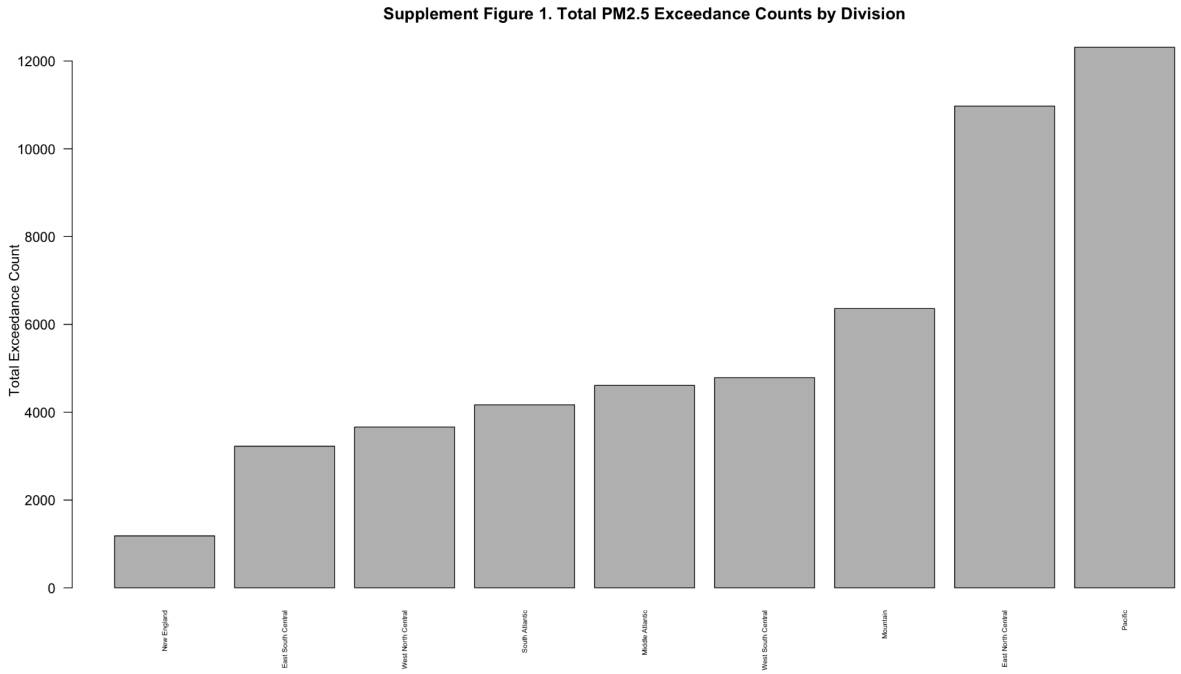
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10 Supplement

Supplement Figure 1. Total PM2.5 Exceedance Counts by Division



Description:

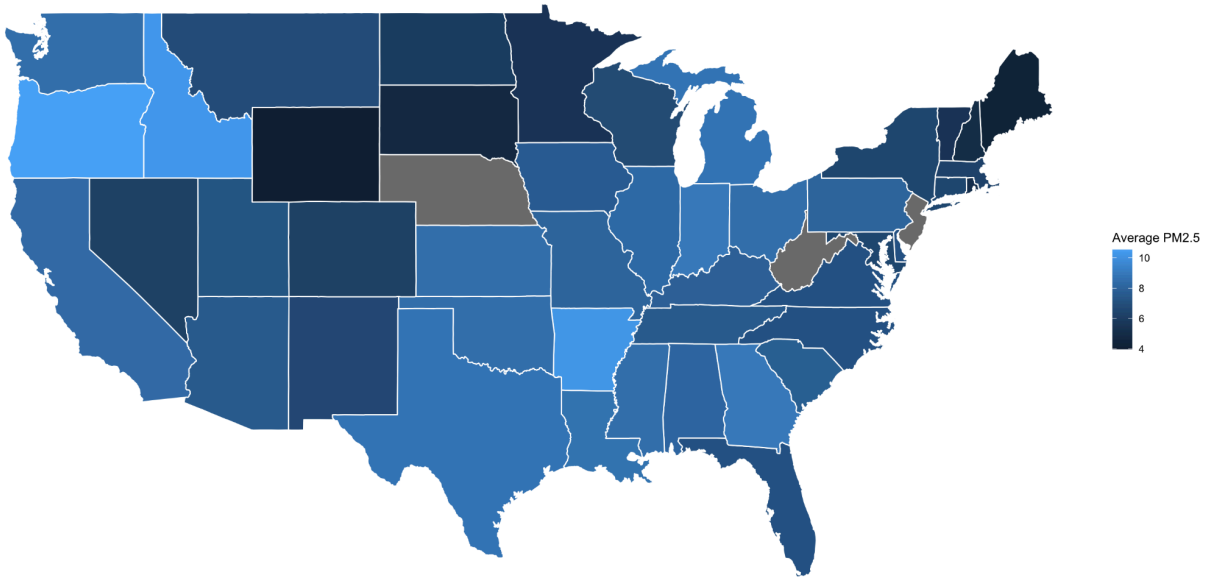
This figure shows the total number of PM2.5 exceedance events across each U.S. Census division. Exceedances were defined as observations where PM2.5 exceeded the WHO threshold of 15 μg/m³. The figure helps visualize how exceedance behavior varies geographically across the United States.

Interpretation:

The Pacific and Mountain divisions showed the highest exceedance counts, supporting patterns observed earlier in the report related to wildfire activity and regional climate conditions. These divisions not only experienced higher average PM2.5 values, but also more frequent extreme pollution events. This figure provides additional support for the count model results by showing that exceedance frequency differs substantially by region. Future work could examine these regions at a finer spatial scale, such as county-level or city-level analysis, to better understand local pollution hotspots and wildfire impacts.

Supplement Figure 2. Average PM2.5 by State

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Description:

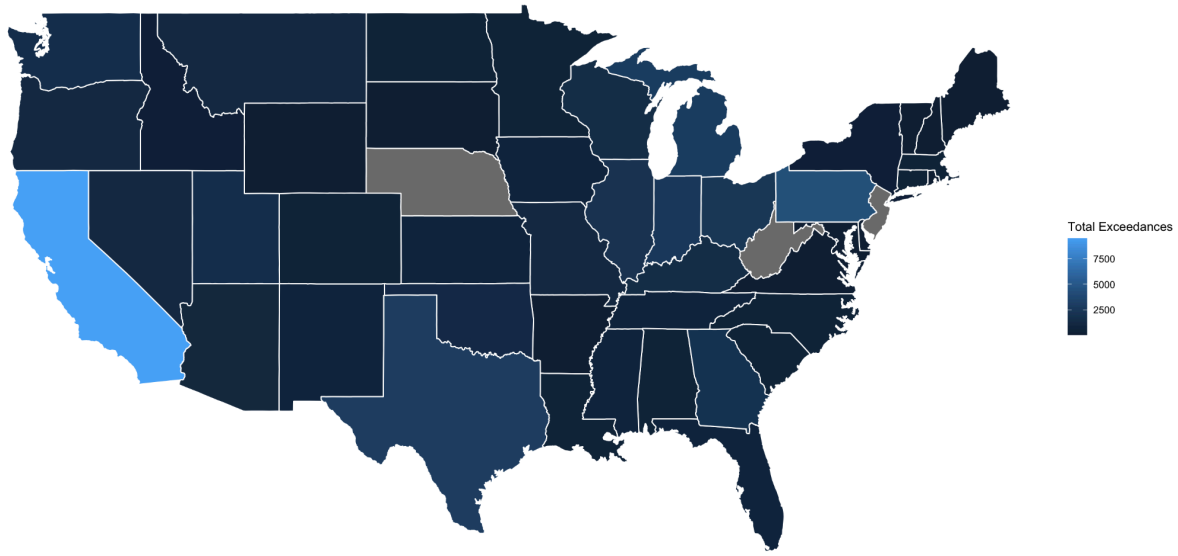
This map displays average PM2.5 concentrations across the United States at the state level. Average PM2.5 values were calculated using aggregated state-day observations from the EPA dataset.

Interpretation:

The western United States generally showed higher average PM2.5 concentrations than eastern regions, particularly in states associated with wildfire activity and dry climate conditions. This map provides more detailed geographic variation than the division-level analyses presented in the main report and helps visualize large-scale spatial pollution patterns. It also shows that several states had limited available observations after preprocessing, likely due to inconsistent environmental reporting or incomplete measurements. Future work could incorporate spatial interpolation methods or additional monitoring datasets to better address missing geographic coverage and improve regional prediction accuracy.

Supplement Figure 3. PM2.5 Exceedance Counts by State

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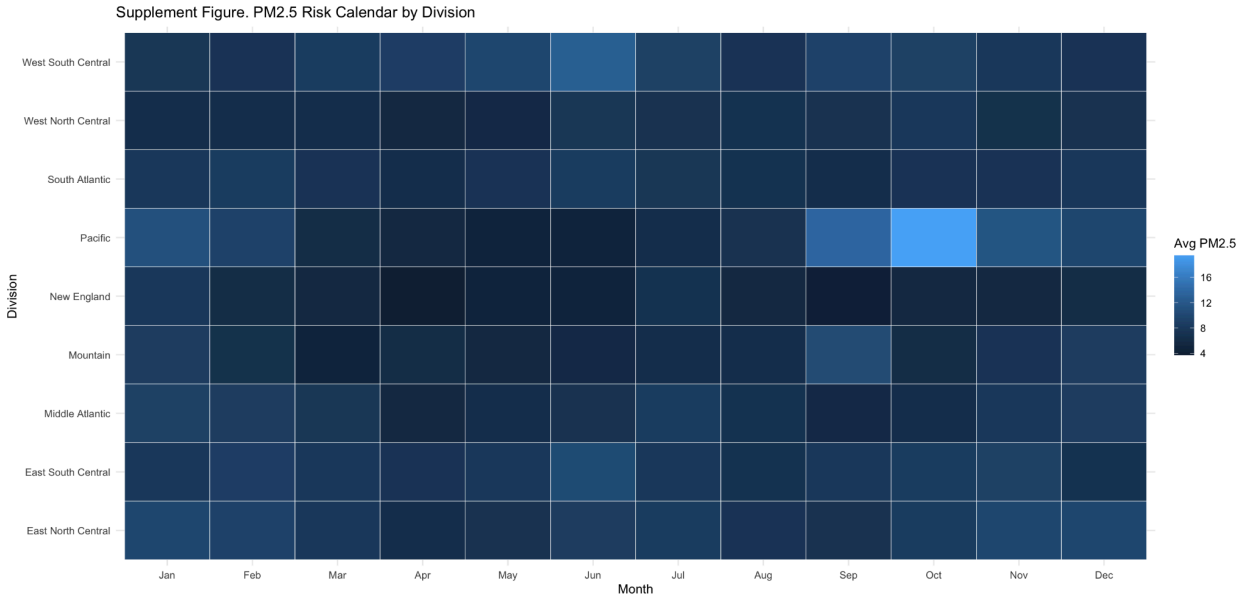
Description:

This map shows the total number of PM2.5 exceedance events for each state using the WHO exceedance threshold of 15 µg/m³.

Interpretation:

States in the western United States experienced noticeably higher exceedance counts, which aligns with wildfire-related pollution patterns discussed throughout the report. The figure demonstrates that exceedance behavior is not evenly distributed across the country and supports the use of geographic predictors in both final models. Future research could compare exceedance patterns between urban and rural regions or evaluate how exceedance frequency changes over multiple years rather than a single study period.

Supplement Figure 4. PM2.5 Risk Calendar by Division



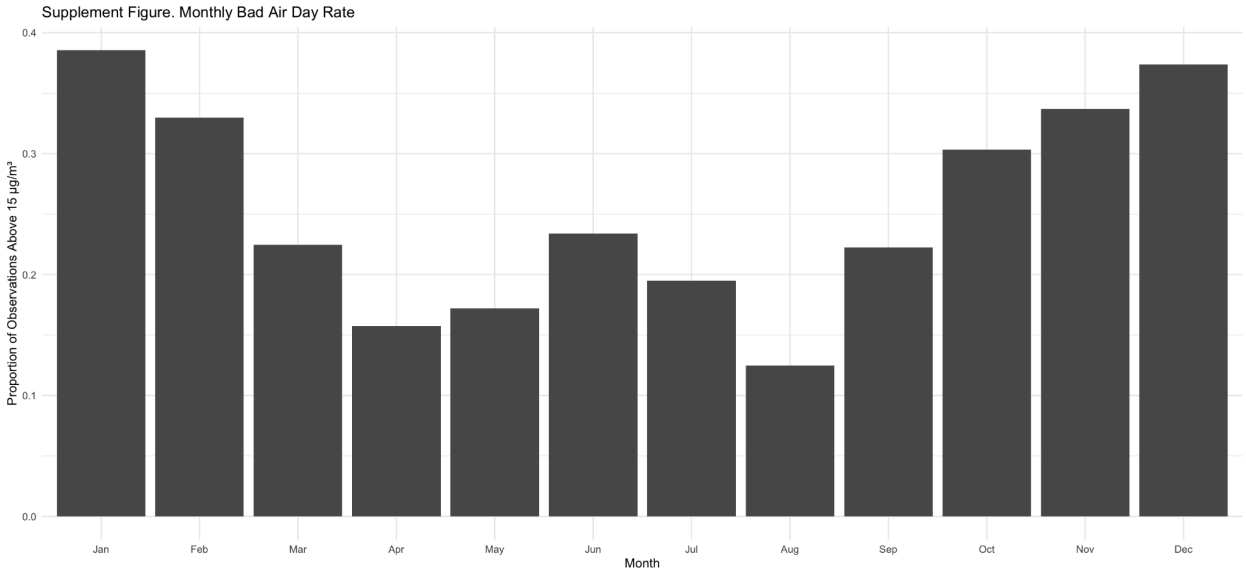
Description:

This heatmap displays average PM2.5 concentrations by month and U.S. Census division. Darker colors represent higher average PM2.5 levels.

Interpretation:

The heatmap highlights strong seasonal patterns in PM2.5 behavior, particularly elevated concentrations during late summer and fall in western divisions. The Pacific division during October stands out as one of the most extreme combinations of time and location, likely reflecting wildfire season conditions. This figure supports the inclusion of both quarter and geographic division in the final models. Future analyses could expand this idea by incorporating additional climate variables such as drought severity, precipitation, or wildfire acreage to better explain seasonal PM2.5 spikes.

Supplement Figure 5. Monthly Bad Air Day Rate



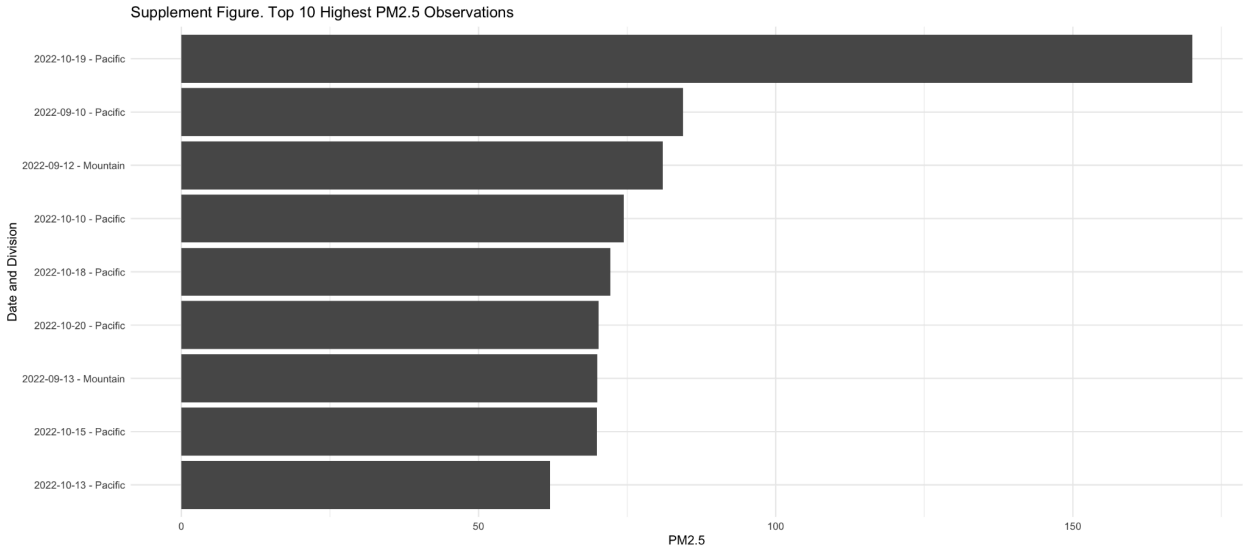
Description:

This figure shows the proportion of observations exceeding the PM2.5 threshold for each month of the year.

Interpretation:

Bad air day rates generally increase during colder months and decrease during the summer, creating a broad U-shaped pattern throughout the year. This suggests that atmospheric conditions, heating-related emissions, and seasonal weather patterns all contribute to PM2.5 exceedance behavior. While wildfire season contributed to large spikes in some western regions, this graph suggests that poor air quality is also influenced by broader seasonal effects across the country. Future work could examine whether these seasonal trends remain consistent over multiple years or under changing climate conditions.

Supplement Figure 6. Top 10 Highest PM2.5 Observations



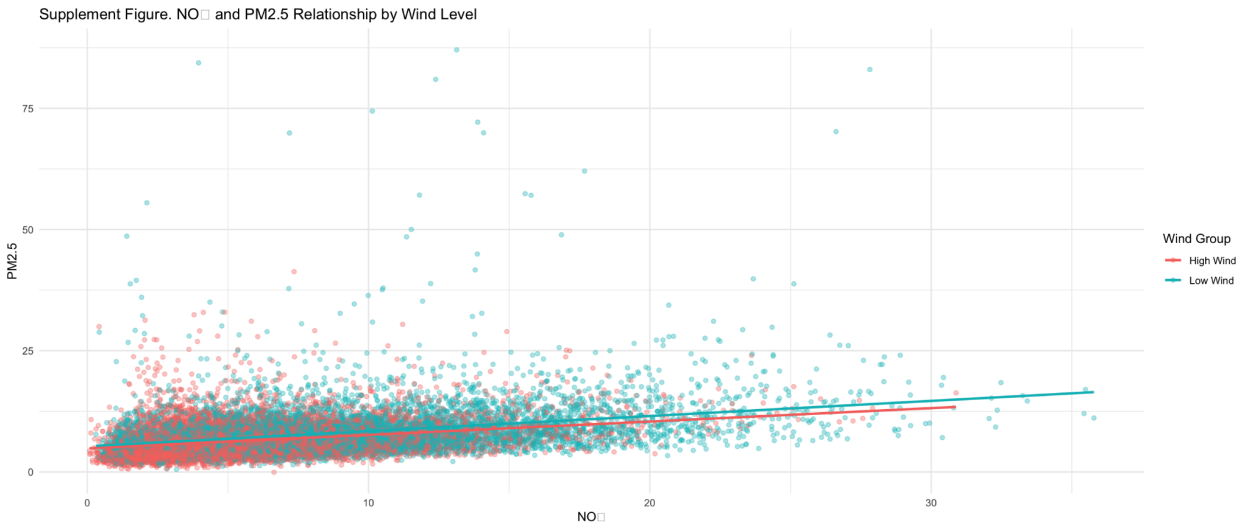
Description:

This figure displays the ten highest PM2.5 observations recorded in the dataset, including their associated dates and locations.

Interpretation:

Extreme PM2.5 events were concentrated within a relatively small number of observations, demonstrating the presence of severe outliers in the data. These events likely correspond to major wildfire episodes or unusual atmospheric conditions. The figure helps explain the strong right-skewness observed in PM2.5 distributions and supports the use of generalized linear models rather than standard linear regression. Future work could focus specifically on extreme-event modeling to better understand the environmental conditions associated with unusually severe pollution episodes.

Supplement Figure 7. NO_2 and PM2.5 Relationship by Wind Level



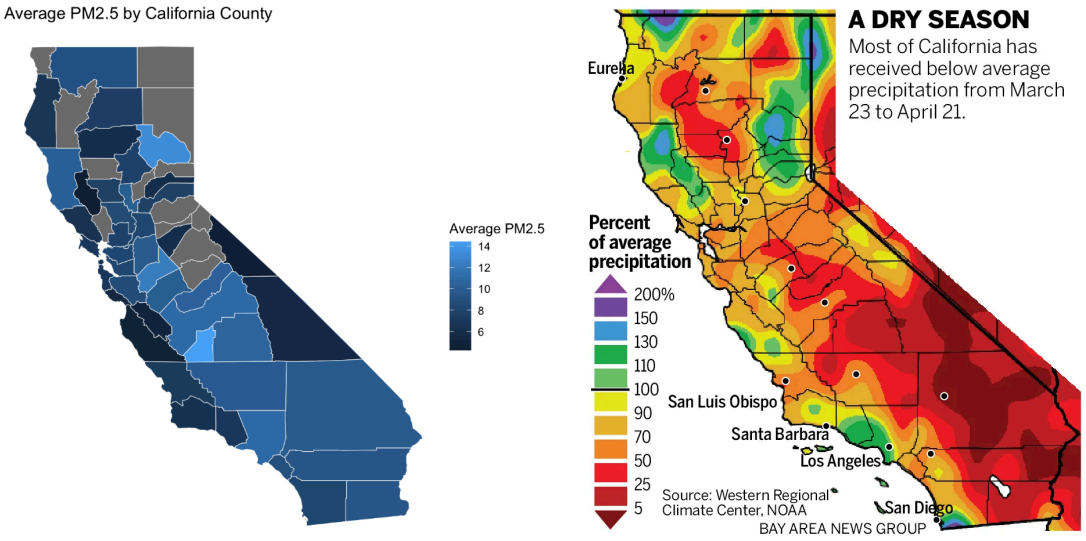
Description:

This scatterplot shows the relationship between NO_2 and PM2.5 under different wind conditions. Observations were grouped into lower and higher wind categories.

Interpretation:

The relationship between NO_2 and PM2.5 appeared stronger during lower wind conditions, while higher wind speeds reduced pollutant buildup through atmospheric dispersion. This visualization supports the inclusion of the NO_2 × windspeed interaction term in the final models because it demonstrates that pollutant effects change depending on atmospheric movement. Future research could incorporate additional atmospheric variables such as pressure systems or wind direction to better model pollutant transport and accumulation.

Supplement Figure 8. Average PM2.5 by California County



Description:

This county-level map displays average PM2.5 concentrations across California counties.

Interpretation:

Higher PM2.5 concentrations were concentrated in inland and southern California regions, while some coastal counties showed lower average levels. This figure provides finer geographic detail than the state-level maps and highlights localized variation within a high-risk state. The accompanying drought map (Monterey Herald) also helps provide environmental context for wildfire-related pollution conditions.